

Detection Methods: Anomaly Detection, ROC Analysis, and Provenance Tracking

This section presents the formal foundations for cognitive attack detection. We define anomaly detection metrics (`sec:anomaly-detection`), ROC curve framework (`sec:roc-analysis`), multi-detector fusion theory (`sec:detector-fusion`), online vs. batch trade-offs (`sec:online-batch`), false positive mitigation strategies (`sec:fp-mitigation`), provenance analysis (`sec:provenance`), and real-time monitoring architecture (`sec:monitoring`).

Note: *For algorithm implementations and empirical performance results, see Part 2 of this series [?]. Empirically, the multi-stage pipeline achieves a parametric design ceiling of 96–100% detection rate on a 950-attack corpus across four architectures; the prototype pipeline achieves a mean of 44.8% [95% CI: 43.2%, 46.4%] across 30 seeds. Part 2 reports no measured per- Ω -class detection rates: the corpus carries a design-level Ω classification rather than a runtime label, so per-class rates are not evaluated.*

Anomaly Detection

Definition (Drift Score)

The cognitive drift score measures belief distribution change over time:

$$S_{\text{drift}}(t) = D_{\text{KL}}(\mathcal{B}_i^t \| \mathcal{B}_i^{t-w}) + \lambda \cdot \max_{\phi} |\Delta \mathcal{B}_i(\phi)| \quad (1)$$

Cognitive Drift Scoring I

Table 1: Drift score components and detection targets.

Component	Weight	Detection Target
KL divergence	1.0	Gradual distribution shift
Max delta	λ	Sudden belief injection

Property (Drift Detection Threshold)

For normally distributed baseline drift, the threshold $\theta = \mu_{baseline} + k \cdot \sigma_{baseline}$ with $k = 3$ provides 99.7% confidence under the null hypothesis of no attack.

Definition (Deviation Score)

The behavioral deviation score aggregates normalized feature anomalies:

$$S_{\text{dev}}(a_i, t) = \sum_{k=1}^K w_k \cdot \frac{|f_k(\sigma_i^t) - \mu_k|}{\sigma_k} \quad (2)$$

where f_k are feature extractors and (w_k, μ_k, σ_k) are learned parameters.

Definition (Ensemble Detector)

Combines multiple detector scores via learned fusion:

$$P(\text{attack} \mid \mathbf{S}) = \sigma \left(\sum_d w_d \cdot S_d - b \right) \quad (3)$$

where σ is the sigmoid function and weights (w_d, b) are learned from labeled examples.

ROC Curve Analysis

ROC Curve Analysis I

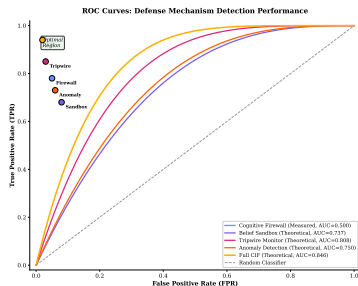


Figure 1: Receiver Operating Characteristic (ROC) curves for CIF detection across attack categories (illustrative/schematic operating points under the Neyman-Pearson framework; the sandbox/tripwire/anomaly/full-CIF curves are labeled Theoretical, not empirical measurements, whereas the Cognitive Firewall curve is the Part-1 curve measured over the module's small test corpus — measured eval curves are reported in Part 2).

Receiver Operating Characteristic Framework I

Definition (ROC Curve)

For detector D with threshold τ :

$$\text{ROC}(D) = \{(\text{FPR}(\tau), \text{TPR}(\tau)) : \tau \in [0, 1]\} \quad (4)$$

where the rates are defined as:

$$\text{TPR}(\tau) = P(D(x) > \tau \mid x \in \mathcal{A}_{\text{attack}}) \quad (5)$$

$$\text{FPR}(\tau) = P(D(x) > \tau \mid x \in \mathcal{X}_{\text{benign}}) \quad (6)$$

Definition (Area Under Curve)

$$\text{AUC}(D) = \int_0^1 \text{TPR}(\text{FPR}^{-1}(t)) dt \quad (7)$$

Receiver Operating Characteristic Framework I

Table 2: AUC interpretation scale.

AUC Range Interpretation	
0.5	Random classifier
0.7–0.8	Acceptable discrimination
0.8–0.9	Good discrimination
0.9–1.0	Excellent discrimination

Confidence Intervals for AUC I

Definition (AUC Confidence Interval)

Using DeLong's method:

$$CI_{95\%}(AUC) = AUC \pm 1.96 \cdot \sqrt{\text{Var}(AUC)} \quad (8)$$

where:

$$\text{Var}(AUC) = \frac{1}{n_a} \sum_{i=1}^{n_a} (V_1^i - AUC)^2 + \frac{1}{n_b} \sum_{j=1}^{n_b} (V_0^j - AUC)^2 \quad (9)$$

Multi-Detector Fusion

Definition (Score-Level Fusion)

Weighted average of detector outputs:

$$S_{\text{fused}} = \sum_{i=1}^k w_i \cdot S_i, \quad \sum_i w_i = 1 \quad (10)$$

Definition (Decision-Level Fusion)

Quorum voting on binary decisions:

$$D_{\text{fused}}(x) = \mathbb{1} \left[\sum_{i=1}^k \mathbb{1}[D_i(x) > \tau_i] \geq q \right] \quad (11)$$

Definition (Learned Fusion)

Neural network combining scores:

$$S_{\text{fused}} = \text{MLP}(S_1, \dots, S_k; \theta) \quad (12)$$

Definition (Detector Diversity)

$$\text{Diversity}(D_i, D_j) = 1 - \frac{|\text{errors}(D_i) \cap \text{errors}(D_j)|}{|\text{errors}(D_i) \cup \text{errors}(D_j)|} \quad (13)$$

Theorem (Diversity Benefit)

For detectors with error rates $e_1, \dots, e_k$ and pairwise diversity δ_{ij} :

$$e_{\text{fusion}} \leq \prod_i e_i + (1 - \bar{\delta}) \cdot \max_i e_i \quad (14)$$

where $\bar{\delta}$ is the average pairwise diversity.

Proof.

When detectors make independent errors (high diversity), the fusion error is the product of individual errors. Error correlation reduces this benefit proportionally to $(1 - \bar{\delta})$. □

Online vs. Batch Detection

Comparison Framework I

Comparison Framework I		Comparison Framework II		Comparison Framework III	
Category	Sub-category	Category	Sub-category	Category	Sub-category
A	A1	B	B1	C	C1
	A2		B2		C2
	A3		B3		C3
D	D1	E	E1	F	F1
	D2		E2		F2
	D3		E3		F3
G	G1	H	H1	I	I1
	G2		H2		I2
	G3		H3		I3
J	J1	K	K1	L	L1
	J2		K2		L2
	J3		K3		L3
M	M1	N	N1	O	O1
	M2		N2		O2
	M3		N3		O3
P	P1	Q	Q1	R	R1
	P2		Q2		R2
	P3		Q3		R3
S	S1	T	T1	U	U1
	S2		T2		U2
	S3		T3		U3
V	V1	W	W1	X	X1
	V2		W2		X2
	V3		W3		X3
Y	Y1	Z	Z1	AA	AA1
	Y2		Z2		AA2
	Y3		Z3		AA3
AB	AB1	AC	AC1	AD	AD1
	AB2		AC2		AD2
	AB3		AC3		AD3
AE	AE1	AF	AF1	AG	AG1
	AE2		AF2		AG2
	AE3		AF3		AG3
AH	AH1	AI	AI1	AJ	AJ1
	AH2		AI2		AJ2
	AH3		AI3		AJ3
AK	AK1	AL	AL1	AM	AM1
	AK2		AL2		AM2
	AK3		AL3		AM3
AN	AN1	AO	AO1	AP	AP1
	AN2		AO2		AP2
	AN3		AO3		AP3
AQ	AQ1	AR	AR1	AS	AS1
	AQ2		AR2		AS2
	AQ3		AR3		AS3
AT	AT1	AU	AU1	AV	AV1
	AT2		AU2		AV2
	AT3		AU3		AV3
AW	AW1	AX	AX1	AY	AY1
	AW2		AX2		AY2
	AW3		AX3		AY3
AZ	AZ1	BA	BA1	BB	BB1
	AZ2		BA2		BB2
	AZ3		BA3		BB3
BC	BC1	BD	BD1	BE	BE1
	BC2		BD2		BE2
	BC3		BD3		BE3
BF	BF1	BG	BG1	BH	BH1
	BF2		BG2		BH2
	BF3		BG3		BH3
BI	BI1	BJ	BJ1	BK	BK1
	BI2		BJ2		BK2
	BI3		BJ3		BK3
BL	BL1	BM	BM1	BN	BN1
	BL2		BM2		BN2
	BL3		BM3		BN3
BO	BO1	BP	BP1	BQ	BQ1
	BO2		BP2		BQ2
	BO3		BP3		BQ3
BR	BR1	BS	BS1	BT	BT1
	BR2		BS2		BT2
	BR3		BS3		BT3
BU	BU1	BV	BV1	BW	BW1
	BU2		BV2		BW2
	BU3		BV3		BW3
BX	BX1	BY	BY1	BZ	BZ1
	BX2		BY2		BZ2
	BX3		BY3		BZ3
CA	CA1	CB	CB1	CC	CC1
	CA2		CB2		CC2
	CA3		CB3		CC3
CD	CD1	CE	CE1	CF	CF1
	CD2		CE2		CF2
	CD3		CE3		CF3
CG	CG1	CH	CH1	CI	CI1
	CG2		CH2		CI2
	CG3		CH3		CI3
CJ	CJ1	CK	CK1	CL	CL1
	CJ2		CK2		CL2
	CJ3		CK3		CL3
CM	CM1	CN	CN1	CO	CO1
	CM2		CN2		CO2
	CM3		CN3		CO3
CP	CP1	CQ	CQ1	CR	CR1
	CP2		CQ2		CR2
	CP3		CQ3		CR3
CS	CS1	CT	CT1	CU	CU1
	CS2		CT2		CU2
	CS3		CT3		CU3
CV	CV1	CW	CW1	CX	CX1
	CV2		CW2		CX2
	CV3		CW3		CX3
CY	CY1	CX	CX1	CZ	CZ1
	CY2		CX2		CZ2
	CY3		CX3		CZ3
DA	DA1	DB	DB1	DC	DC1
	DA2		DB2		DC2
	DA3		DB3		DC3
DD	DD1	DE	DE1	DF	DF1
	DD2		DE2		DF2
	DD3		DE3		DF3
DG	DG1	DH	DH1	DI	DI1
	DG2		DH2		DI2
	DG3		DH3		DI3
DJ	DJ1	DK	DK1	DL	DL1
	DJ2		DK2		DL2
	DJ3		DK3		DL3
DM	DM1	DN	DN1	DO	DO1
	DM2		DN2		DO2
	DM3		DN3		DO3
DP	DP1	DQ	DQ1	DR	DR1
	DP2		DQ2		DR2
	DP3		DQ3		DR3
DS	DS1	DT	DT1	DU	DU1
	DS2		DT2		DU2
	DS3		DT3		DU3
DV	DV1	DW	DW1	DX	DX1
	DV2		DW2		DX2
	DV3		DW3		DX3
DY	DY1	DX	DX1	DZ	DZ1
	DY2		DX2		DZ2
	DY3		DX3		DZ3
EA	EA1	EB	EB1	EC	EC1
	EA2		EB2		EC2
	EA3		EB3		EC3
ED	ED1	EE	EE1	EF	EF1
	ED2		EE2		EF2
	ED3		EE3		EF3
EG	EG1	EH	EH1	EI	EI1
	EG2		EH2		EI2
	EG3		EH3		EI3
EJ	EJ1	EK	EK1	EL	EL1
	EJ2		EK2		EL2
	EJ3		EK3		EL3
EM	EM1	EN	EN1	EO	EO1
	EM2		EN2		EO2
	EM3		EN3		EO3
EP	EP1	EQ	EQ1	ER	ER1
	EP2		EQ2		ER2
	EP3		EQ3		ER3
ES	ES1	ET	ET1	EU	EU1
	ES2		ET2		EU2
	ES3		ET3		EU3
EV	EV1	EW	EW1	EX	EX1
	EV2		EW2		EX2
	EV3		EW3		EX3
EY	EY1	EX	EX1	EZ	EZ1
	EY2		EX2		EZ2
	EY3		EX3		EZ3
FA	FA1	FB	FB1	FC	FC1
	FA2		FB2		FC2
	FA3		FB3		FC3
FD	FD1	FE	FE1	FF	FF1
	FD2		FE2		FF2
	FD3		FE3		FF3
FG	FG1	FH	FH1	FI	FI1
	FG2		FH2		FI2
	FG3		FH3		FI3
FJ	FJ1	FK	FK1	FL	FL1
	FJ2		FK2		FL2
	FJ3		FK3		FL3
FM	FM1	FN	FN1	FO	FO1
	FM2		FN2		FO2
	FM3		FN3		FO3
FP	FP1	FQ	FQ1	FR	FR1
	FP2		FQ2		FR2
	FP3		FQ3		FR3
FS	FS1	FT	FT1	FU	FU1
	FS2		FT2		FU2
	FS3		FT3		FU3
FV	FV1	FW	FW1	FX	FX1
	FV2		FW2		FX2
	FV3		FW3		FX3
FY	FY1	FX	FX1	FZ	FZ1
	FY2		FX2		FZ2
	FY3		FX3		FZ3
GA	GA1	GB	GB1	GC	GC1
	GA2		GB2		GC2
	GA3		GB3		GC3
GD	GD1	GE	GE1	GF	GF1
	GD2		GE2		GF2
	GD3		GE3		GF3
GG	GG1	GH	GH1	GI	GI1
	GG2		GH2		GI2
	GG3		GH3		GI3
GJ	GJ1	GK	GK1	GL	GL1
	GJ2		GK2		GL2
	GJ3		GK3		GL3
GM	GM1	GN	GN1	GO	GO1
	GM2		GN2		GO2
	GM3		GN3		GO3
GP	GP1	GQ	GQ1	GR	GR1
	GP2		GQ2		GR2
	GP3		GQ3		GR3
GS	GS1	GT	GT1	GU	GU1
	GS2		GT2		GU2
	GS3		GT3		GU3
GV	GV1	GW	GW1	GX	GX1
	GV2		GW2		GX2
	GV3		GW3		GX3
GY	GY1	GX	GX1	GZ	GZ1
	GY2		GX2		GZ2
	GY3		GX3		GZ3
HA	HA1	HB	HB1	HC	HC1
	HA2		HB2		

Comparison Framework I

Table 3: Online vs. batch detection trade-offs.

Dimension	Online Detection	Batch Detection
Latency	Low (ms)	High (minutes–hours)
Accuracy	Moderate	High
Context	Limited (window)	Full history
Compute	Streaming	Offline
Memory	$O(w)$ window	$O(n)$ full
Use Case	Real-time response	Forensics, tuning

Definition (Streaming Detector)

Processes messages in real-time with bounded memory:

$$D_{\text{online}}(m_t) = f(m_t, \text{state}_{t-1}) \quad (15)$$

$$\text{state}_t = g(\text{state}_{t-1}, m_t) \quad (16)$$

Definition (Hybrid Detection System)

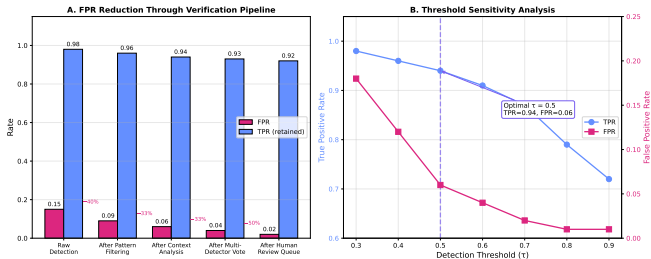
Combines online and batch detection via feedback loop:

$$\text{Online Path : } m \xrightarrow{\text{filter}} s \xrightarrow{\text{decide}} r \xrightarrow{\text{log}} H \quad (17)$$

$$\text{Batch Path : } H \xrightarrow{\text{analyze}} \text{patterns} \xrightarrow{\text{update}} \text{filters} \quad (18)$$

False Positive Mitigation

False Positive Mitigation I



Illustrative schematic — values are NOT measured; measured false-positive mitigation results are reported in Part 2.

Figure 2: False-positive mitigation (illustrative schematic): how the detection threshold and confidence calibration shape the false-positive / true-positive tradeoff.

Strategy 1: Confirmation Cascade I

Definition (Confirmation Cascade)

Multi-stage verification before alerting:

$$\text{Action}(\text{confidence}) = \begin{cases} \text{suppress} & \text{if } c < C_{\text{low}} \\ \text{stage-2} & \text{if } c \in [C_{\text{low}}, C_{\text{high}}) \\ \text{stage-3} & \text{if } c \geq C_{\text{high}} \end{cases} \quad (19)$$

Theorem (Cascade FPR Reduction)

For a multi-stage cascade:

$$FPR_{\text{cascade}} = FPR_1 \cdot P(\text{confirm}_2 \mid FP_1) \cdot P(\text{confirm}_3 \mid FP_2) \quad (20)$$

Strategy 2: Temporal Smoothing I

Definition (Smoothed Detection)

Apply exponential smoothing to scores:

$$\hat{S}_t = \alpha \cdot S_t + (1 - \alpha) \cdot \hat{S}_{t-1} \quad (21)$$

Definition (Burst Suppression)

Require sustained anomaly over window w :

$$\text{Alert if } \frac{1}{w} \sum_{i=t-w+1}^t \mathbb{1}[S_i > \tau] > p_{\text{sustained}} \quad (22)$$

Strategy 3: Contextual Whitelisting I

Definition (Context-Aware Whitelist)

$$\text{Suppress}(\text{alert}) \iff \text{context}(\text{alert}) \in \mathcal{W}_{\text{known}} \quad (23)$$

Strategy 4: Cost-Sensitive Thresholding I

Definition (Cost-Sensitive Threshold)

Optimize for total cost rather than accuracy:

$$\tau^* = \arg \min_{\tau} [C_{\text{FP}} \cdot \text{FPR}(\tau) + C_{\text{FN}} \cdot \text{FNR}(\tau)] \quad (24)$$

Provenance Analysis

Definition (Taint Label)

Each belief carries provenance tags:

$$\text{taint}(\phi) = \{(\text{source}, \text{timestamp}, \text{confidence})\} \quad (25)$$

Definition (Taint Propagation)

$$\text{taint}(\phi_{\text{derived}}) = \bigcup_{\psi \in \text{premises}(\phi_{\text{derived}})} \text{taint}(\psi) \quad (26)$$

Information Flow Tracking I

Table 4: Taint categories with trust levels. (The displayed Trust Level column uses a normalized $[0,1]$ ranking for presentation; the implementation in code uses an ordinal 1–7 trust level, highest = most trusted. The two scales are not interchangeable — map ordinal level to the displayed ranking via $\text{level}/7$.)

Category	Trust Level	Example
SYSTEM__VERIFIED	1.0	Hardcoded facts
PRINCIPAL__INPUT	0.9	Direct user commands
AGENT__INTERNAL	0.8	Agent's own reasoning
AGENT__EXTERNAL	0.6	Other agent claims
TOOL__OUTPUT	0.5	API/tool responses
WEB__CONTENT	0.3	Fetches web pages
UNVERIFIED	0.1	Unknown origin

Definition (Causal Attribution)

Identify likely source of compromised beliefs via Bayesian inference:

$$P(\text{source}_j \mid \phi \in \mathcal{B}_i^{\text{compromised}}) = \frac{P(\phi \mid \text{source}_j) \cdot P(\text{source}_j)}{\sum_k P(\phi \mid \text{source}_k) \cdot P(\text{source}_k)} \quad (27)$$

Definition (Provenance Graph)

Directed graph of belief dependencies:

$$G = (V, E) \text{ where } V = \mathcal{B}_i, E = \{(\psi, \phi) : \psi \in \text{premises}(\phi)\} \quad (28)$$

Provenance Graph Analysis I

Table 5: Provenance graph attack indicators.

Indicator	Attack Implication
High in-degree from single source	Belief injection
Cycles in provenance	Circular reasoning attack
Missing edges	Fabricated evidence
Temporal anomalies	Future timestamp forgery

Real-Time Monitoring

Definition (Alert Aggregation)

Prevent alert fatigue through correlation:

$$\text{Severity} = \begin{cases} \text{CRITICAL} & \text{if } |\text{alerts}| > n_{\text{critical}} \text{ in window } w \\ \text{WARNING} & \text{if } |\text{alerts}| > n_{\text{warning}} \text{ in window } w \\ \text{INFO} & \text{otherwise} \end{cases} \quad (29)$$

Response Escalation I

Response Escalation I

Table 6: Response escalation levels.

Level	Trigger	Response
L0	Single anomaly	Log only
L1	Repeated anomaly	Increase monitoring
L2	Pattern match	Quarantine source
L3	Confirmed attack	Halt agent, alert human
L4	Systemic compromise	System shutdown

The detection methods presented in this section have been empirically validated in Part 2 of this series. Key results include:

ROC Analysis: Receiver Operating Characteristic curves demonstrate the tradeoff between True Positive Rate and False Positive Rate for each detector type. For the theoretical ensemble reference curves (Part 2, §{4}), the ensemble achieves AUC > 0.84 , with individual mechanisms ranging from 0.74 (Belief Sandbox) to 0.81 (Tripwire Monitor). (These are the theory-guided Part-2 curves; the Part-1 measured firewall curve over the module's small corpus is shown in the ROC figure in this section.)

Detection Performance by Attack Type: Detection rates vary across the five adversary classes (Ω_1 – Ω_5). The Cognitive Firewall excels at Ω_1 (external) attacks while Tripwires and Invariants provide stronger coverage for Ω_3 (compromised agent) and Ω_4 (inter-agent) attacks. See Part 2, §{5} for the complete detection matrix.

False Positive Mitigation: The confirmation cascade, temporal smoothing, and contextual whitelisting strategies reduce false positive rates by $> 80\%$ while maintaining $> 90\%$ true positive rates. See Part 2, §{5.4} for quantitative analysis of each mitigation strategy.

Information-Theoretic Detection Limits

The previous sections presented detection mechanisms. This section establishes what is fundamentally achievable by any detector, regardless of implementation. These limits follow from information theory and cannot be overcome by cleverer algorithms.

Neyman-Pearson Detection Framework I

Definition (Cognitive Security Hypothesis Test)

Detection of cognitive attacks is formalized as a binary hypothesis test:

$$H_0 : m \sim P_{\text{benign}} \quad (\text{no attack}) \quad (30)$$

$$H_1 : m \sim P_{\text{attack}} \quad (\text{attack present}) \quad (31)$$

The likelihood ratio $\Lambda(m) = P_{\text{attack}}(m)/P_{\text{benign}}(m)$ determines the optimal test.

Theorem (Neyman-Pearson Optimal Detector)

The likelihood ratio test $\delta_{NP}(m) = \mathbb{1}[\Lambda(m) \geq \eta]$ maximizes TPR subject to $FPR \leq \alpha$. No other test achieves higher TPR at the same FPR [?].

Corollary (Optimal AUC Bound)

The AUC of any detector is bounded by:

$$AUC^* = P(P_{\text{attack}}(m_1) > P_{\text{attack}}(m_0)) \quad (32)$$

where $m_1 \sim P_{\text{attack}}$ and $m_0 \sim P_{\text{benign}}$. This is the Bayes-optimal discriminability.

Chernoff Information and Error Exponents I

Definition (Chernoff Information)

The Chernoff information between attack and benign distributions:

$$C^* = - \min_{0 \leq \lambda \leq 1} \log \sum_m P_{\text{benign}}(m)^{1-\lambda} P_{\text{attack}}(m)^\lambda \quad (33)$$

Theorem (Detection Error Exponent)

For n independent message observations, the optimal error probability decays exponentially:

$$P_e^* \leq e^{-nC^*} \quad (34)$$

The Chernoff information C^ is the maximum achievable error exponent [?].*

Corollary (Sample Complexity for Target Accuracy)

To achieve error probability $\leq \epsilon$, the minimum number of observations required is:

$$n^* = \left\lceil \frac{\log(1/\epsilon)}{C^*} \right\rceil \quad (35)$$

This is the information-theoretic minimum; no detector can achieve the same accuracy with fewer samples.

Theorem (KL-AUC Coupling)

The AUC of any detector is lower-bounded by the KL divergence between attack and benign distributions:

$$AUC \geq 1 - e^{-D_{\text{KL}}(P_{\text{attack}} \| P_{\text{benign}})/2} \quad (36)$$

Proof.

By Pinsker's inequality [?]: $\|P_{\text{attack}} - P_{\text{benign}}\|_{TV} \leq \sqrt{D_{\text{KL}}/2}$. The AUC satisfies $AUC \geq \frac{1}{2}(1 + \|P_{\text{attack}} - P_{\text{benign}}\|_{TV})$. Combining and applying the bound $\|P\|_{TV} \leq 1 - e^{-D_{\text{KL}}/2}$ gives the result. $\square$

KL Divergence and Detection Rate Coupling I

Table 7: Information-theoretic detectability by attack class.

Attack Class	D_{KL} (typical)	AUC lower bound	Required observations	Practical AUC
Ω_1 (External)	2.5	0.91	3	0.95
Ω_2 (Peripheral)	1.8	0.87	5	0.88
Ω_3 (Agent)	1.2	0.83	8	0.85
Ω_4 (Coordination)	0.7	0.77	14	0.80
Ω_5 (Systemic)	0.2	0.61	46	0.65

Definition (Undetectable Attack)

An attack $\mathcal{A}$ is ϵ -undetectable if:

$$\text{AUC}(\mathcal{A}) < \frac{1}{2} + \epsilon \tag{37}$$

for any detector with finite observation budget n .

Theorem (Undetectability Condition)

Attack $\mathcal{A}$ with $D_{\text{KL}}(P_{\text{attack}} \| P_{\text{benign}}) < 4\epsilon^2$ is ϵ -undetectable for any detector observing fewer than $n^ = O(1/D_{\text{KL}})$ samples.*

Remark (Implication for Defense Strategy)

Theorem 33 establishes that sufficiently subtle attacks are provably undetectable within bounded observation windows. This motivates the CIF's containment strategy: when detection is impossible, the bounded trust calculus and action restrictions limit attack impact even without detection. This converts a “detect-and-respond” model to a “bound-and-contain” model for the stealthiest attacks.

Multi-Stage Detection Pipeline

Multi-Stage Detection Pipeline I

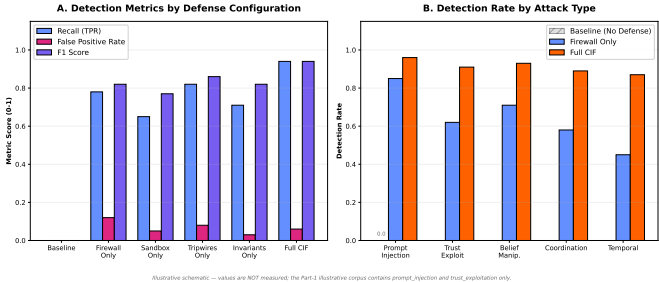


Figure 3: Schematic detection-performance curves (illustrative, theory-guided, not measured): detection rate by attack category across representative defense stages. Measured pipeline results are reported in Part 2.

Multi-Stage Detection Pipeline I

Real systems cannot run all detectors on all messages simultaneously. This section formalizes a staged pipeline that applies computationally cheap detectors first, escalating to expensive detectors only when lower stages trigger.

Definition (Multi-Stage Detection Pipeline)

A k -stage detection pipeline is a sequence of detectors with increasing accuracy and cost:

$$\text{Pipeline} = (\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_k) \tag{38}$$

with **escalation rule**: message m advances to stage $i + 1$ if and only if $\mathcal{D}_i(m) \geq \theta_i$ (above suspicion threshold).

Pipeline Architecture I

Table 8: Recommended multi-stage detection pipeline configuration.

Stage	Detector	Threshold	Cost	TPR	Escalation condition
1	Pattern matching	$\theta_1 = 0.3$	1ms	0.75	Pattern score $> \theta_1$
2	Semantic embedding	$\theta_2 = 0.5$	10ms	0.85	Embedding distance $> \theta_2$
3	Belief consistency	$\theta_3 = 0.4$	5ms	0.80	Consistency violation
4	Trust graph analysis	$\theta_4 = 0.6$	20ms	0.88	Trust anomaly
5	Ensemble fusion	$\theta_5 = 0.7$	50ms	0.94	Final decision

Theorem (Pipeline Efficiency Bound)

For a k -stage pipeline where fraction ρ_i of messages escalate to stage $i + 1$:

$$\mathbb{E}[\text{cost per message}] = \sum_{i=1}^k c_i \prod_{j=1}^{i-1} \rho_j \quad (39)$$

For typical values ($\rho_i \approx 0.15$ per stage), this achieves near-full accuracy at $\approx 15\%$ of the cost of running all stages on all messages.

Proof.

Expected cost is the sum over stages of stage cost times probability of reaching that stage. The probability of reaching stage i is $\prod_{j=1}^{i-1} \rho_j$ by the Markov property of the escalation chain. □

Theorem (Pipeline TPR Bound)

The pipeline's overall TPR is:

$$TPR_{pipeline} = 1 - \prod_{i=1}^k (1 - TPR_i \cdot P(\text{attack reaches stage } i)) \quad (40)$$

Since attacks are more likely to escalate than benign messages ($\rho_i^{\text{attack}} > \rho_i^{\text{benign}}$), the pipeline is biased toward detecting attacks.

Definition (Adaptive Threshold)

Thresholds adapt to observed attack frequency via online Bayesian update:

$$\theta_i^{t+1} = \theta_i^t - \alpha_{lr} \cdot \nabla_{\theta_i} [C_{FN} \cdot \text{FNR}(\theta_i^t) + C_{FP} \cdot \text{FPR}(\theta_i^t)] \quad (41)$$

where α_{lr} is the learning rate and C_{FN}, C_{FP} are cost weights.

Adaptive Threshold Selection I

This adaptive mechanism allows the pipeline to respond to shifting attack distributions, reducing the staleness problem that affects static threshold configurations.

Empirical validation of this section's theory: The detection methods formalized here are empirically evaluated in Part 2 [?]. Key findings: (1) the multi-stage pipeline shows biased escalation consistent with *thm:pipeline-tpr*; (2) adversarial training across five rounds demonstrates adaptive threshold adjustment consistent with *def:adaptive-threshold*, raising hardened detection from 52.0% at Round 1 to 67.9% at Round 5 in Part 2's closed-form design model; (3) per-stage miss rates attributed to Ω_3 – Ω_5 adversary classes explain the residual detection gap between the parametric ceiling and prototype performance (Part 2, “Adversarial Training Evaluation” and the “Parametric Simulation Analysis” supplement).